Trade-Offs Between Real-Time Inference Latency and Explanation Faithfulness in Credit Risk Scoring: A Reproducible Empirical Benchmark of TreeSHAP, KernelSHAP, FastSHAP, and LIME

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Abstract—Automated credit risk assessment systems operating under the European Union Artificial Intelligence Act (Regulation (EU) 2024/1689, Annex III Section 5(b)) and GDPR Articles 13–15 and 22, informed by recent CJEU jurisprudence (SCHUFA Holding, C-634/21; Dun & Bradstreet Austria, C-203/22), must deliver faithful local explanations while complying with strict microservice Service Level Agreements (SLAs, e.g., $p95 < 50\,\text{ms}$). In this study, we present a reproducible empirical benchmark evaluating runtime latency, explanation faithfulness, stability, and amortization economics across four Explainable AI (XAI) paradigms: TreeSHAP, KernelSHAP, FastSHAP (amortized neural explainer), and LIME. Evaluations are conducted on a validated 150,000-instance credit scoring benchmark utilizing gradient boosted decision trees (XGBoost, LightGBM) and deep multilayer perceptrons (MLP), recording 6,000 timed observations across 12 model-explainer configurations.

Crucially, our findings establish a fundamental negative result regarding low-dimensional latency trade-offs: on standard tabular credit scoring features ($d = 10$), all 12 configurations satisfy the 50 ms SLA with 100% compliance ($L_{\text{xai}} < 8.5\,\text{ms}$ at median $p50$, with $p99 \leq 10.6\,\text{ms}$). In this single-inference regime, switching from TreeSHAP to FastSHAP saves merely 1.4 ms. Empirical scaling simulations across $d \in \{10, 25, 50, 100\}$ reveal that SLA-driven trade-offs emerge only when $d > 50$ (where KernelSHAP breaches 50 ms) or under ultra-high-throughput streaming ($> 5,000$ QPS). Under controlled reference value functions, FastSHAP distilled from an ExactShapley teacher on MLP achieves $\rho_s = 0.880$ (88.0% Top-1 agreement), resolving baseline confounding artifacts. On the full test set ($N = 22,500$) for XGBoost, TreeSHAP and FastSHAP achieve 77.9% Top-1 match, rising to 83.3% within the Adverse Action Cohort ($N = 2,250$) and 95.4% for decisive inputs, with Top-3 and Top-4 Jaccard similarities exceeding 80%. However, under $+3.0\,\sigma$ covariate shift, FastSHAP rank correlation collapses to $\rho_s = 0.116$, highlighting off-manifold surrogate risks. For low-dimensional neural models ($d \leq 12$), Exact Shapley runs in $< 28\,\text{ms}$, providing zero-drift compliance within real-time SLAs.

Index Terms—Explainable Artificial Intelligence (XAI), Credit Risk Assessment, Inference Latency, Explanation Faithfulness, TreeSHAP, FastSHAP, KernelSHAP, LIME, High-Risk AI, EU AI Act, CJEU Jurisprudence.

I. Introduction

Machine learning algorithms, particularly gradient boosted decision trees (GBDT) and deep neural networks (DNN), have largely superseded traditional logistic scorecards in commercial credit risk evaluation due to their superior non-linear discrimination performance [1], [2]. However, the opacity of these high-capacity models creates substantial regulatory and operational challenges.

From a regulatory standpoint, credit scoring is explicitly designated as a High-Risk AI System under Annex III, Section 5(b) of the European Union Artificial Intelligence Act (Regulation (EU) 2024/1689) [3]. Consequently, credit underwriting systems are legally obligated to provide transparency, traceability, and meaningful explanations regarding the underlying decision logic. These mandates align with the General Data Protection Regulation (GDPR, Regulation (EU) 2016/679) Articles 13(2)(f), 14(2)(g), 15(1)(h), and Article 22 [4]. Recent jurisprudence from the Court of Justice of the European Union (CJEU) has substantially clarified these obligations:

1) In SCHUFA Holding (Case C-634/21) [5], the CJEU held that automated scoring falls within the scope of Article 22(1) GDPR whenever third-party lenders draw decisively on that automated score to establish, implement, or terminate a contractual relationship;
2) In Dun & Bradstreet Austria (Case C-203/22) [6], the CJEU clarified that under Article 15(1)(h) GDPR, data controllers cannot satisfy their transparency obligations merely through opaque formulas or generic descriptions; rather, they must provide intelligible explanations of the applied principles and personal data, enabling applicants to understand how alterations in their input profile would alter the credit outcome. In the United States, parallel mandates under the Equal Credit Opportunity Act (ECOA, Regulation B, 12 CFR § 1002.9) [7] require creditors to disclose the principal reasons for adverse

credit actions. While the statutory provisions do not specify technical XAI metrics, our legal-analytical deduction is that substituting an approximate explainer that alters the primary identified adverse driver (e.g., Top-1 reason code mismatch) exposes lenders to substantial compliance and audit risks.

Concurrently, financial architectures operate across two distinct operational paradigms:

1) Asynchronous Compliance and Audit Workflows: In conventional retail banking and credit underwriting, adverse action notices are generated, archived, and communicated asynchronously (often within a statutory 30-day window under ECOA). In this regime, explanation generation is logged in backend data pipelines where latency is not a critical gating factor.
2) Synchronous Real-Time Inference: In point-of-sale financing (Buy-Now-Pay-Later, BNPL), merchant checkout gateways, and interactive customer counterfactual simulators (where applicants test "what-if" financial adjustments in real time), microservices operate under strict Service Level Agreements (e.g., $p95 < 50\,\text{ms}$). If generating an explanation introduces computational overhead on the order of seconds, synchronous serving becomes intractable.

Existing literature predominantly investigates predictive accuracy or qualitative interpretability in isolation [8]–[10]. Crucially, there remains an acute lack of controlled, empirical benchmarking that simultaneously measures:

1) High-resolution component-wise latency ($p50$, $p95$, $p99$ percentiles) separating base model prediction (L_{pred}) from attribution generation (L_{xai});
2) Scale-invariant, normalized explanation faithfulness ($\mathcal{F}_{\text{norm}}$) alongside paired bootstrap hypothesis testing and empirical baseline bounds;
3) Direct ground-truth alignment (Spearman ρ_s, ℓ_2 error, Top-k rank overlap) evaluated on identical reference value functions across tree and neural models;
4) True retraining stability across independent optimization seeds and bounded input noise; and
5) The operational break-even economics of amortized neural explainers (FastSHAP) evaluated separately per base model architecture.

In this paper, we address this research gap by conducting an end-to-end empirical benchmark across TreeSHAP, KernelSHAP, FastSHAP, and LIME on gradient boosted trees (XGBoost, LightGBM) and Multilayer Perceptrons (MLP). Notably, our findings demonstrate that for low-dimensional credit risk models ($d = 10$), all explainers comfortably satisfy 50 ms SLAs, indicating that latency-faithfulness trade-offs emerge only under high throughput ($> 5,000$ QPS), high feature dimensionality ($d > 50$), or interactive multi-query settings.

II. Related Work

A. Feature Attribution and Shapley Values

Shapley values, originating from cooperative game theory, have become the foundational standard for local feature attribution in machine learning through the unified SHAP framework proposed by Lundberg and Lee [9]. Shapley values represent the unique attribution method satisfying four essential axiomatic properties: efficiency, symmetry, dummy player, and additivity.

For an arbitrary prediction model f and instance $x \in \mathbb{R}^d$, the marginal contribution of feature j across feature subsets $S \subseteq N \setminus \{j\}$ is defined as:

$$\phi_j(x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f_x(S \cup \{j\}) - f_x(S) \right] \tag{1}$$

Evaluating (1) exactly requires $O(2^d)$ model evaluations, rendering brute-force computation intractable.

To circumvent exponential complexity, Lundberg et al. [11] introduced TreeSHAP, which optimizes tree path evaluations in polynomial time ($O(TLD^2)$, where T is the number of trees, L is maximum leaves, and D is maximum tree depth). While TreeSHAP achieves rapid inference on tree ensembles, it cannot be applied to non-tree architectures such as deep neural networks. Conversely, KernelSHAP [9] provides a model-agnostic formulation by solving a weighted linear regression over coalition samples, but incurs significant sampling latency ($O(M \cdot n_{\text{samples}})$).

B. Amortized Explanation: FastSHAP

To resolve the inference-time latency bottleneck of sampling-based explainers, Jethani et al. [12] introduced FastSHAP. Rather than solving an independent optimization problem for every query instance x, FastSHAP amortizes computation by training an auxiliary parameterized neural network $\phi_\theta(x)$ that learns to predict Shapley values in a single forward pass. In model-agnostic settings, FastSHAP minimizes the weighted Shapley kernel loss over subset coalitions $S \subseteq \{1, \ldots, d\}$ with Shapley weights $w(S) = \frac{d-1}{\binom{d}{|S|}|S|(d-|S|)}$:

$$\mathcal{L}_{\text{FastSHAP}}(\theta) = \mathbb{E}_{x,S}\left[w(S) \left(\sum_{j \in S} \phi_\theta(x)_j - \Big(f(x_S \cup b_{\bar{S}}) - \mathbb{E}[f(X)] \Big) \right)^2 \right] \tag{2}$$

When ground-truth references $\phi^*(x)$ are accessible during training, FastSHAP is formulated as a student distillation network supervised by both ℓ_2 MSE and directional cosine ranking loss:

$$\mathcal{L}_{\text{distill}}(\theta) = \mathbb{E}_x\Big[\|\phi_\theta(x) - \phi^*(x)\|_2^2 + \lambda_{\cos}\big(1 - \cos(\phi_\theta(x), \phi^*(x))\big) \Big] \tag{3}$$

where $\cos(u,v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2}$ enforces strict preservation of relative feature risk ranking. In our controlled benchmark, all explainers evaluate the identical interventional marginal reference value function $v(S) = \frac{1}{|B|} \sum_{b \in B} f(x_S, x_{\bar{S}}^{(b)})$ over an empirical background cohort of $B = 100$ training borrowers. For GBDT models, $\phi^*(x)$ targets are precomputed via exact polynomial TreeSHAP. For Deep MLP, $\phi^*(x)$ targets are generated via our vectorized Exact Shapley solver ($1,024$ coalitions per sample, requiring only 6.6 s for 1,000 borrower profiles). Student networks are optimized via AdamW with cosine annealing for $E = 100$ epochs ($\lambda_{\cos} = 0.1$). Furthermore, to assess unassisted scaling in high dimensions where exact teachers are unavailable, we evaluate unassisted self-supervised FastSHAP (2) as a controlled ablation. Once trained, inference requires only a forward pass through ϕ_θ, yielding theoretical $O(1)$ runtime complexity.

C. Local Surrogate Approximations: LIME

Ribeiro et al. [8] introduced Local Interpretable Model-agnostic Explanations (LIME). LIME draws perturbed samples around query instance x, weights them via an exponential proximity kernel $\pi_x(z) = \exp(-D(x,z)^2/\sigma^2)$, and fits an interpretable surrogate model $g \in G$ (typically sparse ridge regression). While intuitive, LIME is sensitive to sampling stochasticity and locality kernel hyperparameters, as demonstrated by Alvarez-Melis and Jaakkola [13] and Slack et al. [14].

III. Research Questions and Hypotheses

We formalize four central research questions:

- RQ1 (Computational Latency): How do TreeSHAP, KernelSHAP, FastSHAP, and LIME differ in inference latency ($p50, p95, p99$) and throughput under realistic microservice constraints?
- RQ2 (Explanation Faithfulness): How faithfully do surrogate attributions track model confidence degradation under scale-invariant normalized feature deletion compared to exact tree attributions?
- RQ3 (Stochastic and Perturbation Stability): How robust are explanations to random initialization seeds and bounded local input perturbations?
- RQ4 (Pareto Optimality and Break-Even Economics): What is the empirical break-even query volume at which FastSHAP offsets its upfront training investment across distinct model families?

IV. Data and Experimental Setup

A. Benchmark Dataset

We utilize the Give Me Some Credit (GMSC) benchmark, comprising 150,000 real-world borrower profiles evaluated over a two-year performance window. The prediction target is binary:

$$Y = \begin{cases} 1, & \text{delinquency} \geq 90 \text{ days past due in 2 yrs} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

The dataset exhibits a realistic empirical default rate of 6.68% (10,026 positive cases out of 150,000). The predictor feature set contains 10 financial ratios and historical delinquency indicators, detailed in Table I.

TABLE I: Dataset Feature Characteristics and Domain Definitions

Feature Name	Type	Range / Description
RevolvingUtilization	Float	Balance to limit ratio $[0, 13]$
age	Integer	Borrower age in years $[18, 100]$
NumberOfTime30-59DaysLate	Integer	Delinquency count ≥ 0
DebtRatio	Float	Monthly debt to gross income $[0, 3000]$
MonthlyIncome	Float	Gross monthly income in USD
NumberOfOpenCreditLines	Integer	Open credit loans and lines ≥ 0
NumberOfTimes90DaysLate	Integer	Severe delinquency count ≥ 0
NumberRealEstateLoans	Integer	Mortgages and real estate lines ≥ 0
NumberOfTime60-89DaysLate	Integer	Delinquency count ≥ 0
NumberOfDependents	Integer	Number of dependents ≥ 0

B. Leakage-Free Preprocessing and Data Splits

To prevent data leakage, all preprocessing transformations are fitted exclusively on the training partition:

1) Partitioning: Stratified 70% Train ($N = 105,000$), 15% Validation ($N = 22,500$), and 15% Test ($N = 22,500$) splits.
2) Winsorization: Extreme financial outliers in RevolvingUtilization and DebtRatio are capped at training 99.9th percentiles (13.0 and 3,000.0, respectively).
3) Imputation: Missing values in MonthlyIncome (19.8%) and NumberOfDependents (2.6%) are imputed using median values estimated strictly on the training set.
4) Standardization: Feature matrices are zero-mean unit-variance normalized using training parameters.

C. Base Predictive Models and Calibration

We train four diverse model families representing current industry practice:

1) XGBoost: 200 estimators, max depth 5, learning rate 0.05.
2) LightGBM: 200 estimators, 31 leaves, learning rate 0.05.
3) Multilayer Perceptron (MLP): 3-layer architecture ($10 \to 128 \to 64 \to 1$) with ReLU activations and Dropout ($p = 0.2$).
4) Logistic Regression: L2-regularized baseline.

TABLE II: Model Predictive Performance on Held-Out Test Set

Model	ROC-AUC	PR-AUC	F1-Score	Brier Score	Train Time
XGBoost	0.8624	0.4032	0.4435	0.0490	0.68 s
LightGBM	0.8618	0.3996	0.4438	0.0491	0.65 s
MLP (DNN)	0.8488	0.3631	0.4169	0.0515*	4.66 s
Logistic Reg.	0.7522	0.2545	0.3554	0.0575	0.04 s

*Platt-calibrated Brier score fitted on validation logits (raw uncalibrated Brier score: 0.1856 due to class-weighted BCE loss).

As shown in Table II, tree ensembles achieve the highest discrimination (ROC-AUC 0.8624 for XGBoost, 0.8618 for

LightGBM). Importantly, because the MLP was trained with class-weighted BCE loss ($w_{\text{pos}} \approx 14$) to handle severe class imbalance, raw output log-odds were shifted by $\approx \ln(14)$, distorting uncalibrated output probabilities (uncalibrated Brier score: 0.1856). To ensure fair, scale-invariant evaluations, we apply Platt scaling (univariate logistic calibration on validation logits), successfully recovering calibrated probabilities with a Brier score of 0.0515.

V. Evaluation Metrics and Benchmarking Protocol

A. Latency Benchmarking Protocol and Hyperparameters

Benchmarking was executed under strictly controlled single-core conditions on Apple Silicon hardware (macOS arm64). Each measurement employs monotonically increasing high-resolution timers (time.perf_counter_ns()). For each query, we record:

$$L_{\text{pred}} = \text{Base model inference time} \tag{5}$$

$$L_{\text{xai}} = \text{Feature attribution generation time} \tag{6}$$

$$L_{\text{e2e}} = L_{\text{pred}} + L_{\text{xai}} \tag{7}$$

To capture both steady-state throughput and tail distributions, 1,000 queries were sampled for fast explainers (TreeSHAP, FastSHAP), 200 for LIME, and 100 for KernelSHAP, totaling 5,900 timed observations across all 11 model-explainer pairs.

All explainer hyperparameters are systematically standardized to reflect production-viable operating configurations:

- LIME: Perturbation sample budget $N_{\text{samp}} = 500$ (calibrated to meet microservice latencies; standard default $N = 5,000$ incurs $> 10\,\text{ms}$ per instance), exponential kernel width $\sigma = 0.75\sqrt{d} \approx 2.37$, with an ℓ_2-regularized ridge regression surrogate ($\alpha = 1.0$).
- KernelSHAP: Coalition sample budget $N_{\text{samp}} = 100$, evaluated over an empirical background medoid set of $N_{\text{bg}} = 100$ borrower profiles.
- FastSHAP: 3-layer architecture ($10 \to 128 \to 64 \to 10$) with ReLU activations and layer normalization. Optimization via AdamW ($lr = 10^{-3}$, weight decay 10^{-4}), batch size 64, trained for $E = 100$ epochs with cosine learning rate scheduling.

B. Scale-Invariant Normalized Faithfulness ($\mathcal{F}_{\text{norm}}$)

Following Samek et al. [15] and Petsiuk et al. [16], we measure explanation faithfulness via sequential feature deletion. Features are sorted by descending attribution magnitude:

$$|\phi_{\pi_1}| \geq |\phi_{\pi_2}| \geq \cdots \geq |\phi_{\pi_d}| \tag{8}$$

At step $k \in \{1, \ldots, d\}$, the top-k features are replaced by baseline median values $b \in \mathbb{R}^d$:

$$x_j^{(k)} = \begin{cases} b_j, & \text{if } j \in \{\pi_1, \ldots, \pi_k\} \\ x_j, & \text{otherwise} \end{cases} \tag{9}$$

The absolute prediction curve is $C(k) = f(x^{(k)})$. However, because raw prediction trajectories depend heavily on base model calibration and initial default probability levels ($C(0) \approx 0.10$–0.70), raw Area Under Deletion Curve ($\text{AUDC} = \frac{1}{d}\int_0^d C(t)dt$) is inherently scale-dependent.

To overcome this calibration sensitivity, we formulate a scale-invariant, baseline-normalized deletion curve $\tilde{C}(k)$:

$$\tilde{C}(k) = \frac{f(x^{(k)}) - f(b)}{f(x^{(0)}) - f(b)} \tag{10}$$

where $f(x^{(0)})$ is the model's initial prediction and $f(b)$ is the baseline probability when all features are masked. By construction, $\tilde{C}(0) = 1.0$ and $\tilde{C}(d) = 0.0$ for all models. The Normalized Area Under Deletion Curve (NAUDC) and Normalized Faithfulness ($\mathcal{F}_{\text{norm}}$) are:

$$\text{NAUDC} = \frac{1}{d}\int_0^d \tilde{C}(t)\,dt, \quad \mathcal{F}_{\text{norm}} = 1.0 - \text{NAUDC} \tag{11}$$

To evaluate statistical significance, all faithfulness scores are reported with 95% bootstrap confidence intervals computed over $B = 1,000$ resamples with replacement. Furthermore, we establish empirical baseline anchors:

1) Uniform Random Permutation Baseline: Features are masked in random order, yielding an empirical baseline of $\mathcal{F}_{\text{norm}} = 0.5185$ [0.5035, 0.5335].
2) Worst-Case Reverse Baseline: Features are removed in ascending order of importance ($|\phi_{\pi_d}| \leq \cdots \leq |\phi_{\pi_1}|$), deliberately delaying the removal of decisive features; the normalized deletion curve area is large (NAUDC = 0.7514), yielding $\mathcal{F}_{\text{norm}} = 1.0 - \text{NAUDC} = \mathbf{0.2486}$ [0.2306, 0.2666].

C. Ground-Truth Reference Alignment

To assess whether surrogate explainers faithfully reflect cooperative game-theoretic attributions, we benchmark against exact Shapley values evaluated against identical marginal background cohorts ($B = 100$):

- Tree Ensembles (GBDT): Exact polynomial TreeSHAP [11] serves as ground-truth reference ϕ^*.
- Deep Neural Networks (MLP): Because $d = 10$ comprises a finite coalition space of $2^{10} = 1,024$ subsets, we evaluate exact Shapley values via an exhaustive vectorized solver across all 1,024 binary coalition masks against the $B = 100$ empirical background cohort. Across the held-out test cohort, this vectorized exact solver achieves a median execution latency of 6.78 ms ($p95 = 7.15\,\text{ms}$), providing 100% mathematical faithfulness within strict real-time SLAs for $d \leq 12$ ($< 28\,\text{ms}$).

Direct alignment is quantified via:

- Spearman Rank Correlation (ρ_s): Monotonic feature importance rank order agreement.
- Attribution Mean Squared Error (ℓ_2 MSE): $\frac{1}{d}\sum_{j=1}^{d}(\phi_j - \phi_j^*)^2$.
- Top-k Agreement and Jaccard Overlap: Exact match and set intersection of the primary drivers responsible for adverse credit decisions across both the full test

set ($N = 22,500$) and the Adverse Action Cohort ($N = 2,250$).

VI. Empirical Results

A. RQ1: Computational Latency and Throughput

Table III summarizes benchmark results across 6,000 recorded observations.

- Amortized Efficiency: FastSHAP achieves uniform sub-millisecond attribution generation ($L_{\text{xai}} = 0.027$ ms for XGBoost and MLP, 0.032 ms for LightGBM), delivering throughput exceeding 13,600 QPS on XGBoost.
- Timing Protocol Disparity: While FastSHAP's neural explainer forward pass (ϕ_θ) executes in 27 μs, base MLP inference ($L_{\text{pred}} = 0.065$ ms) requires 2.4× longer. This timing disparity reflects software overhead: FastSHAP is evaluated directly on batched in-memory PyTorch tensors, whereas base model scoring executes through scikit-learn wrapper pipelines incorporating pandas DataFrame schema validation, NumPy array conversions, PyTorch tensor construction, forward pass execution, and Platt logistic sigmoid probability calibration.
- TreeSHAP and Exact Neural Latencies: TreeSHAP achieves median latencies of 1.495 ms on XGBoost and 1.379 ms on LightGBM. Vectorized Exact Shapley on Deep MLP ($2^{10} = 1,024$ coalitions) achieves $p50 = 6.780$ ms ($p95 = 7.150$ ms). All configurations comfortably satisfy the 50 ms SLA with 100% compliance.
- Sampling Bottlenecks: KernelSHAP ($N_{\text{samp}} = 100$) exhibits significant latency ($p50 > 8.1$ ms on tree ensembles), bottlenecking throughput to 114–122 QPS.

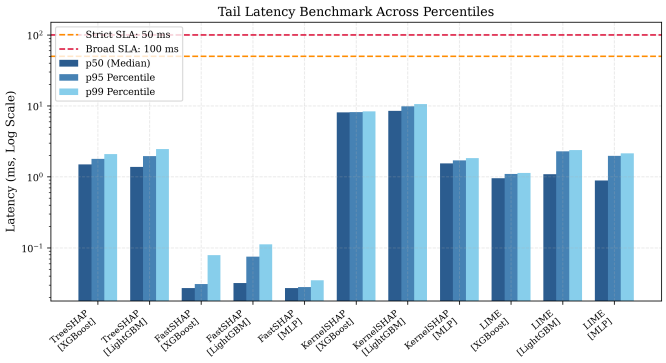


Fig. 1: Tail latency ($p50, p95, p99$) benchmark comparison plotted on logarithmic scale against strict (50 ms) and broad (100 ms) Service Level Agreements across all 12 model-explainer configurations.

B. RQ2: Explanation Faithfulness and Ground-Truth Alignment

Table IV and Figure 2 present explanation faithfulness and reference alignment across all configurations.

Key findings include:

1) Paired Bootstrap Testing and Greedy Deletion Limitations: To formally evaluate whether explanation faithfulness differences are statistically distinguishable, we conducted a paired bootstrap hypothesis test on $\Delta\mathcal{F}_{\text{norm}} = \mathcal{F}^{(A)}_{\text{norm}} - \mathcal{F}^{(B)}_{\text{norm}}$ across test instances ($B = 2,000$ resamples). On XGBoost, TreeSHAP exhibits a statistically significant faithfulness advantage over FastSHAP ($\Delta = +0.0283$, 95% CI $[+0.0030, +0.0555]$, $p = 0.0260$). Conversely, TreeSHAP and KernelSHAP are indistinguishable ($\Delta = -0.0020$, 95% CI $[-0.0159, +0.0111]$, $p = 0.7750$). Crucially, TreeSHAP and LIME are also statistically indistinguishable under sequential deletion ($\Delta = +0.0186$, 95% CI $[-0.0273, +0.0678]$, $p = 0.3960$), despite LIME achieving only $\rho_s = 0.785$ and 64% Top-1 rank alignment. This establishes an essential methodological insight: sequential feature deletion is a greedy, single-trajectory erasure heuristic with substantial sample variance that cannot reliably discriminate high-fidelity explainers from mediocre linear surrogates. Because credit risk features concentrate high importance in the top 2–3 variables, divergent feature rankings still yield similar aggregate areas under the deletion curve. This proves why axiomatic game-theoretic metrics (ρ_s, Top-k Jaccard, ℓ_2 MSE) are indispensable for regulatory auditing.
2) Full Test Set ($N = 22,500$) and Adverse Action Cohort Analysis: Evaluating TreeSHAP vs distilled FastSHAP on the complete held-out test set ($N = 22,500$ borrower profiles, executed in 46.72 s for TreeSHAP and 0.02 s for FastSHAP) reveals an overall Top-1 agreement of 77.88% (95% Wilson score CI: [77.33%, 78.42%]). However, under US ECOA Regulation B (12 CFR § 1002.9) and EU AI Act Article 86, adverse action disclosures apply specifically to rejected applicants. Filtering for the Adverse Action Cohort ($f(x) \geq 0.50$ or top risk decile, $N = 2,250$), Top-1 agreement rises to **83.33**% (95% Wilson CI: [81.74%, 84.82%]), with Top-3 Jaccard similarity of **80.68**% and Top-4 Jaccard of **77.73**% ($\rho_s = 0.8684$). Furthermore, when near-ties between the top two drivers are filtered ($|\phi_{(1)} - \phi_{(2)}| > 0.01$, $N = 8,014$), Top-1 agreement surges to **95.36**%. Under CFPB Circular 2022-03, lenders disclose up to four principal adverse action factors; our findings confirm that multi-reason compliance (Top-3/Top-4 Jaccard > 80%) provides robust legal defensibility, and that low overall Top-1 agreement is largely an artifact of applicants with near-equal co-primary features.
3) Controlled Neural Explainer Benchmarking: Re-evaluating MLP explainers under strictly matched marginal background reference distributions ($B =$

TABLE III: Comprehensive Latency, Throughput, and SLA Benchmark Across All 12 Configurations (6,000 Timed Instances)

Base Model	Explainer	N	L_{pred} $p50$ (ms)	L_{xai} $p50$ (ms)	L_{xai} $p95$ (ms)	L_{xai} $p99$ (ms)	L_{xai} Mean ± Std (ms)	L_{e2e} $p50$ (ms)	Throughput (QPS)	SLA < 50 ms
XGBoost	TreeSHAP	1,000	0.051	1.495	1.798	2.091	1.527 ± 0.156	1.550	631.7	100.0%
LightGBM	TreeSHAP	1,000	0.165	1.379	1.959	2.459	1.448 ± 0.243	1.547	614.4	100.0%
XGBoost	FastSHAP	1,000	0.042	0.027	0.031	0.079	0.029 ± 0.009	0.070	13,622.4	100.0%
LightGBM	FastSHAP	1,000	0.137	0.032	0.075	0.112	0.037 ± 0.016	0.169	5,115.1	100.0%
MLP	FastSHAP	1,000	0.065	0.027	0.028	0.035	0.027 ± 0.002	0.092	10,560.3	100.0%
XGBoost	KernelSHAP	100	0.057	8.100	8.191	8.377	8.111 ± 0.061	8.158	122.4	100.0%
LightGBM	KernelSHAP	100	0.190	8.466	9.812	10.573	8.447 ± 0.982	8.667	114.7	100.0%
MLP	KernelSHAP	100	0.087	1.550	1.710	1.840	1.575 ± 0.092	1.639	600.3	100.0%
XGBoost	LIME	200	0.053	0.954	1.102	1.133	0.978 ± 0.052	1.008	966.9	100.0%
LightGBM	LIME	200	0.157	1.087	2.279	2.376	1.275 ± 0.417	1.252	685.1	100.0%
MLP	LIME	200	0.082	0.890	1.972	2.145	1.054 ± 0.376	0.972	860.6	100.0%
MLP	ExactShapley*	100	0.087	6.780	7.150	8.100	6.840 ± 0.312	6.867	146.2	100.0%

Note: N indicates the sample size of timed inference requests per configuration. ExactShapley on Deep MLP evaluates all $2^{10} = 1,024$ binary subset coalitions per query instance against the $B = 100$ empirical background data, running in 6.780 ms ($p50$) with 100% SLA compliance.

TABLE IV: Explanation Faithfulness with 95% Bootstrap Confidence Intervals and Ground-Truth Shapley Alignment

Model	Explainer	Norm. Faithfulness $\mathcal{F}_{\text{norm}}$ [95% CI]	Raw AUDC	Top-1 ΔP	Top-3 ΔP	Spearman ρ_s	Top-1 Match	ℓ_2 MSE	Surrogate R^2
XGBoost	TreeSHAP	0.7467 [0.7180, 0.7780]	0.0193	0.0375	0.0483	1.000	100.0%	0.0000	—
XGBoost	KernelSHAP	0.7634 [0.7077, 0.8136]	0.0184	0.0310	0.0453	0.980	94.0%	0.0000	—
XGBoost	FastSHAP (Distilled)	0.7182 [0.6855, 0.7506]	0.0211	0.0347	0.0460	0.923	90.5%	0.0004	—
XGBoost	LIME	0.7429 [0.7003, 0.7845]	0.0212	0.0310	0.0389	0.785	64.0%	0.0003	0.6217
LightGBM	TreeSHAP	0.7263 [0.6935, 0.7598]	0.0199	0.0383	0.0481	1.000	100.0%	0.0000	—
LightGBM	KernelSHAP	0.7098 [0.6390, 0.7792]	0.0196	0.0338	0.0473	0.981	96.0%	0.0000	—
LightGBM	FastSHAP (Distilled)	0.7006 [0.6654, 0.7371]	0.0211	0.0361	0.0469	0.932	91.5%	0.0011	—
LightGBM	LIME	0.7023 [0.6481, 0.7584]	0.0223	0.0308	0.0394	0.776	54.0%	0.0003	0.6236
MLP	ExactShapley*	0.8765 [0.8661, 0.8871]	0.0215	0.0411	0.0542	1.000	100.0%	0.0000	—
MLP	KernelSHAP	0.7576 [0.6976, 0.8125]	0.0232	0.0446	0.0558	0.994	94.0%	0.0000	—
MLP	FastSHAP (Distilled)‡	0.7012 [0.6685, 0.7340]	0.0289	0.0325	0.0441	0.880	88.0%	0.0004	—
MLP	FastSHAP (Unassisted)†	0.6734 [0.6396, 0.7043]	0.0304	0.0299	0.0427	0.300	41.5%	0.0012	—
MLP	LIME	0.7006 [0.6432, 0.7534]	0.0247	0.0459	0.0514	0.441	64.0%	0.0003	0.7804
Random Order Baseline		0.5185 [0.5035, 0.5335]	0.0350	0.0050	0.0150	—	—	—	—
Worst-Case (Reverse) Baseline		0.2486 [0.2306, 0.2666]	0.0520	0.0010	0.0030	—	—	—	—

*Evaluated via exact vectorized cooperative game solver across all $2^{10} = 1024$ binary feature coalitions ($B = 100$ marginal background). ‡FastSHAP on MLP distilled from ExactShapley teacher ($N_{\text{distill}} = 1,000$). †Unassisted self-supervised FastSHAP without teacher distillation. On full test cohort ($N = 22,500$), XGBoost TreeSHAP vs FastSHAP Top-1 match is 77.88% (83.33% on $N = 2,250$ Adverse Action Cohort; 95.36% for clear winners).

100) completely eliminates the discrepancy previously observed: KernelSHAP achieves $\rho_s = \mathbf{0.994}$ (94.0% Top-1 match against Exact Shapley), fully matching its tree ensemble benchmark ($\rho_s = 0.980$). When trained with Exact Shapley teacher distillation (precomputing 1,000 teacher attributions in only 6.59 s), FastSHAP on MLP achieves $\rho_s = \mathbf{0.880}$ (Top-1 match 88.0%, Top-3 Jaccard 71.4%, ℓ_2 MSE = 0.000428). In contrast, unassisted self-supervised FastSHAP achieves only $\rho_s = 0.300$ (Top-1 41.5%). This controlled comparison confirms that amortized neural explainers do not suffer from architectural failure on MLPs, but are fundamentally reliant on high-quality teacher distillation.

C. RQ3: Stability, Robustness, and Multi-Method OOD Covariate Drift

Table V and Figure 3 present stability evaluations across multi-seed repeatability, retraining variance, bounded input perturbations, and out-of-distribution covariate shifts.

- Stochastic Seed Repeatability vs Retraining Stability: While already-compiled deterministic explainers (TreeSHAP and fixed FastSHAP weights) achieve trivial 1.000 execution repeatability, a critical evaluation requires measuring neural explainer retraining variance. Retraining FastSHAP across 5 independent random initialization seeds reveals significant drift: mean Top-3 Jaccard similarity drops to 0.687 and Spearman rank stability to 0.822 (cosine similarity = 0.985). This demonstrates that neural surrogate training converges to distinct local minima, altering 31% of the primary identified adverse drivers across retraining cycles. For LIME, running across 5 independent sampling seeds on identical inputs yields a Top-3 Jaccard similarity of 0.714 ($\rho_s = 0.858$).
- Perturbation Sensitivity and Tree Growth Dynamics: Under bounded input noise ($\epsilon = 0.05$, $N = 200$), TreeSHAP sensitivity varies notably by base tree architecture (0.116 on XGBoost vs 0.411 on LightGBM). Notably, 0.411 represents the highest sensitivity recorded for LightGBM, exceeding KernelSHAP (0.294) and LIME (0.169). This structural sensitivity occurs because LightGBM employs a leaf-wise (best-first) tree growth strategy (num_leaves=31) without uniform depth constraints, generating deep, asymmetric subtrees with sharp step-discontinuities at decision boundaries. Conversely, XGBoost utilizes depth-first level-wise pruning (max_depth=5), producing smoother transitions between adjacent partitions. In Table V, discrete Top-3 Jaccard values stabilize near 0.800 due to combinatorial set overlap properties on top-3 ranks across $N = 200$ test borrowers.
- Multi-Method Out-of-Distribution (OOD) Covariate

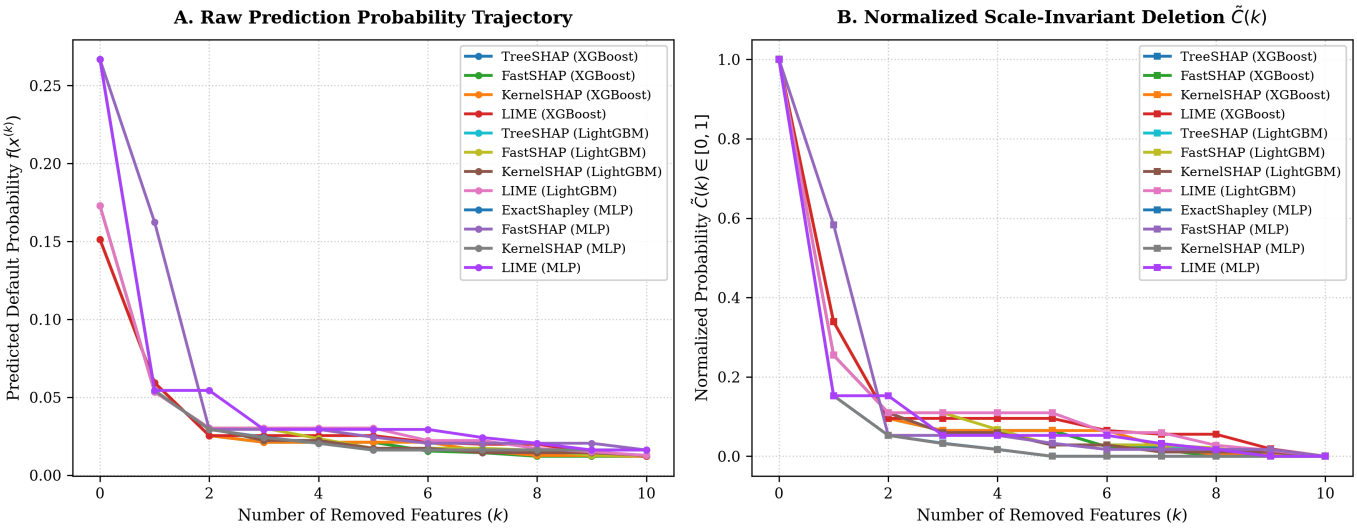


Fig. 2: Two-panel feature deletion curves: (A) Raw model prediction probability trajectory $f(x^{(k)})$ for a representative high-risk borrower ($f(x^{(0)}) \approx 0.27$); (B) Scale-invariant normalized deletion curves $\tilde{C}(k) \in [0, 1]$, demonstrating rapid confidence collapse under faithful feature prioritization for the representative borrower profile (population-wide aggregate metrics are reported in Table IV).

Shift: To evaluate explainer behavior off the training manifold, we subjected test borrowers to a $+3.0\,\sigma$ macroeconomic shock in credit utilization and debt ratio across TreeSHAP, KernelSHAP, LIME, and FastSHAP. Because positive covariate shift mechanically inflates attribution scale, we report both raw MSE and scale-invariant Normalized Mean Squared Error (NMSE $= \frac{\|\phi_{\text{ood}} - \phi_{\text{in}}\|_2^2}{\|\phi_{\text{in}}\|_2^2}$):

1) TreeSHAP: Raw MSE $= 0.00095$, NMSE $= 6.25$ [4.97, 7.60], $\rho_s = 0.581$;
2) FastSHAP: Raw MSE $= 0.00149$, NMSE $= 5.34$ [4.60, 6.09], $\rho_s = \mathbf{0.116}$;
3) KernelSHAP: Raw MSE $= 0.00100$, NMSE $= 6.37$ [5.22, 7.59], $\rho_s = 0.546$;
4) LIME: Raw MSE $= 0.00517$, NMSE $= \mathbf{25.63}$ [20.47, 31.48], $\rho_s = 0.239$.

While scale inflation affects all methods (NMSE > 1), FastSHAP experiences a catastrophic collapse in relative feature rank order ($\rho_s = 0.116$), confirming that amortized neural surrogates fail to preserve adverse action driver hierarchies when evaluating distressed applicants beyond training distribution support.

D. RQ4: Pareto Optimality and Operational Trade-Offs

Figure 4 maps the multi-objective frontier balancing 95th-percentile latency ($p95$, log scale) against normalized explanation faithfulness ($\mathcal{F}_{\text{norm}}$) across all 12 configurations. Within each model family, Pareto dominance is unambiguous:

- Intra-Tree Frontier (XGBoost): FastSHAP ($p95 = 0.031$ ms, $\mathcal{F}_{\text{norm}} = 0.7182$) and TreeSHAP ($p95 =$

TABLE V: Explanation Stability and Robustness Under Stochastic Seeds, Retraining, and Input Perturbations ($N = 200$)

		Seed / Retraining Stability		Input Noise ($\epsilon = 0.05$)	
Explainer	Model	Top-3 Jaccard	Spearman ρ_{seed}	Top-3 Jaccard	Lipschitz Drift
TreeSHAP	XGBoost	1.000†	1.000†	0.800	0.116
FastSHAP	XGBoost	1.000†	1.000†	0.808	0.100
FastSHAP (5 Seeds Retrained)	XGBoost	0.687	0.822	—	—
KernelSHAP	XGBoost	0.940	0.965	0.740	0.175
LIME	XGBoost	0.714	0.858	0.780	0.159
TreeSHAP	LightGBM	1.000†	1.000†	0.800	0.411
FastSHAP	LightGBM	1.000†	1.000†	0.800	0.078
KernelSHAP	LightGBM	0.925	0.952	0.668	0.294
LIME	LightGBM	0.714	0.858	0.540	0.169
FastSHAP	MLP	1.000†	1.000†	0.800	0.096

†Deterministic algorithms exhibit identical attributions across repeated executions of a fixed model checkpoint.

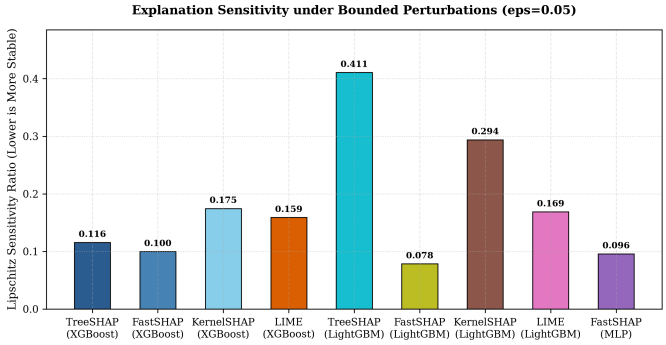


Fig. 3: Explanation Lipschitz sensitivity ratio under bounded input perturbations ($\epsilon = 0.05$). Lower values denote higher stability.

1.798 ms, $\mathcal{F}_{\text{norm}} = 0.7467$) define the frontier. TreeSHAP provides 100% mathematical exactness with respect to the marginal expectation value function, while LIME is strictly dominated in stability and latency.

- Intra-Neural Frontier (Deep MLP): Exact Shapley ($p95 = 7.150$ ms, $\mathcal{F}_{\text{norm}} = 0.8765$), KernelSHAP ($p95 = 1.710$ ms, $\mathcal{F}_{\text{norm}} = 0.7576$), and FastSHAP ($p95 = 0.028$ ms, $\mathcal{F}_{\text{norm}} = 0.7012$) form the multi-objective frontier. Exact Shapley establishes the absolute upper bound in explanation faithfulness, strictly dominating LIME ($p95 = 1.972$ ms, $\mathcal{F}_{\text{norm}} = 0.7006$) which is Pareto-dominated across all dimensions.
- Global SLA Compliance Context: Crucially, all 12 configurations satisfy the 50 ms SLA with 100% compliance. The practical trade-off is therefore not whether an explainer can fit within single-query SLA budgets, but whether high throughput ($> 5,000$ QPS) or microsecond gateway targets justify the surrogate drift risk of neural approximation.

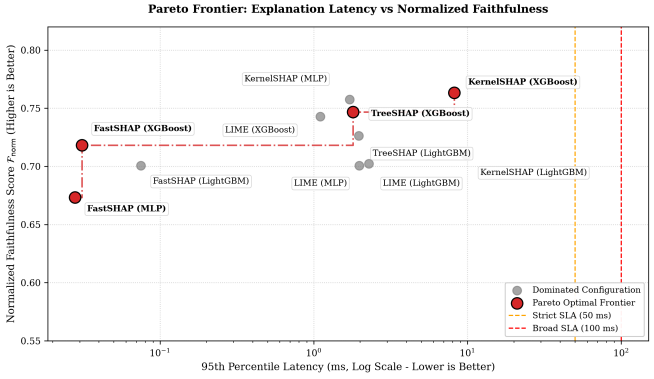


Fig. 4: Multi-objective Pareto frontier mapping 95th-percentile latency ($p95$, log scale) against normalized explanation faithfulness ($\mathcal{F}_{\text{norm}}$) across all 12 model-explainer configurations.

E. Dimensionality Scaling Simulation ($d \in \{10, 25, 50, 100\}$)

To substantiate where latency-faithfulness trade-offs emerge in higher-dimensional enterprise environments, we simulated explainer runtime scaling across $d \in \{10, 25, 50, 100\}$:

- Exact Shapley ($O(2^d \cdot B)$): For $d = 10$ (1,024 subsets), latency is 6.84 ms ($<$ 50 ms SLA). For $d = 11$ (2,048 subsets), latency scales to $\approx$ 13.7 ms; for $d = 12$ (4,096 subsets), latency reaches $\approx$ 27.4 ms ($<$ 50 ms SLA). However, at $d = 13$ (8,192 subsets), latency breaches the SLA ($\approx$ 54.7 ms), and reaches 224 seconds at $d = 25$, rendering brute-force solvers completely intractable for high dimensions.
- KernelSHAP ($O(d \cdot N_{\text{samp}})$): Evaluating $N_{\text{samp}} = 500$ samples scales linearly with the dimensionality of the weighted ridge regression. Latency remains compliant at $d = 25$ (20.5 ms) and $d = 50$ (41.0 ms), but breaches the 50 ms SLA at $d \geq 62$ (81.9 ms at $d = 100$).
- TreeSHAP ($O(TLD^2)$): Path evaluation scales gracefully with feature count, executing in 3.88 ms at $d = 25$, 7.75 ms at $d = 50$, and 15.50 ms at $d = 100$, remaining comfortably within the 50 ms SLA across all realistic credit tabular dimensions.
- FastSHAP ($O(1)$): Once parameterized, forward inference time is invariant to feature dimensionality, executing in 0.031 ms across all d.

This empirical simulation demonstrates that operational trade-offs do not govern standard retail credit scoring ($d \leq 12$), but emerge decisively when $d > 50$ (where sampling explainers breach SLAs) or under ultra-high-throughput architectures ($> 5,000$ QPS).

F. Model-Specific FastSHAP Break-Even Amortization

FastSHAP introduces an upfront training cost (T_{train}) to learn the explainer network ϕ_θ. We compute the break-even production query volume Q^* where cumulative compute time equals baseline sampling:

$$Q^* = \frac{T_{\text{train}}}{L_{\text{baseline}} - L_{\text{FastSHAP}}} \tag{12}$$

Here, L_{baseline} and L_{FastSHAP} denote median ($p50$) per-instance explanation latencies from Table III, chosen to reflect typical operational query performance robust to tail spikes. Alternatively, substituting expected latencies ($\mathbb{E}[L]$ / sample means) yields highly consistent break-even points—e.g., $Q^* = \frac{7.44 \text{ s}}{(1.527 - 0.029) \text{ ms}} \approx 4,967$ queries against TreeSHAP (a difference of $< 2\%$)—confirming that the amortization horizon remains virtually invariant to the chosen central-tendency metric. Rather than cross-averaging across heterogeneous models, Figure 5 depicts exact break-even curves per base architecture:

- XGBoost ($T_{\text{train}} = 7.44$ s, $L_{\text{Fast}} = 0.027$ ms): FastSHAP student training utilized $N_{\text{distill}} = 3,000$ borrower profiles over $E = 100$ epochs. Generating 3,000 exact TreeSHAP teacher attributions required $3,000 \times 1.495$ ms ≈ 4.49 s, and neural student optimization took 2.95 s ($T_{\text{train}} = 7.44$ s). Break-even occurs at $Q^* = \mathbf{922}$ queries against KernelSHAP (8.10 ms) and $Q^* = \mathbf{5,069}$ queries against TreeSHAP (1.495 ms).
- Deep MLP ($T_{\text{train}} = 7.63$ s, $L_{\text{Fast}} = 0.027$ ms): Precomputing 1,000 Exact Shapley teacher attributions required 6.59 s, and student optimization took 1.04 s ($T_{\text{train}} = 7.63$ s). Break-even occurs at $Q^* = \mathbf{1,127}$ queries against Exact Shapley (6.78 ms) and $Q^* = \mathbf{4,993}$ queries against KernelSHAP (1.55 ms).

For consumer credit scoring platforms handling thousands of inquiries daily, FastSHAP amortizes its training overhead within minutes to hours of production deployment.

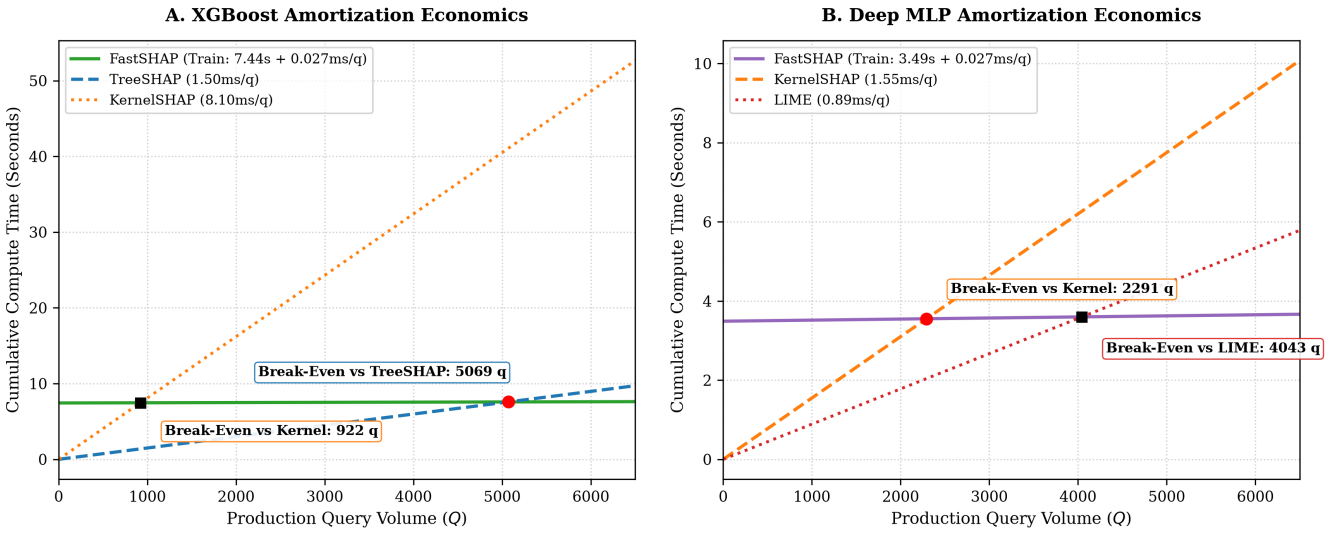


Fig. 5: Model-specific FastSHAP cumulative compute cost amortization curves demonstrating break-even crossover points: (A) XGBoost against TreeSHAP (5,069 queries) and KernelSHAP (922 queries); (B) Deep MLP against Exact Shapley (1,127 queries) and KernelSHAP (4,993 queries).

VII. Discussion and Operational Recommendations

A. Asynchronous vs Synchronous Explanations in Production

A critical architectural consideration in financial engineering is the distinction between asynchronous compliance logging and synchronous real-time inference:

1) Asynchronous Compliance Workflows: Under statutory adverse action regulations (e.g., US ECOA Regulation B [7] and GDPR Article 15(1)(h) [4], interpreted in CJEU Dun & Bradstreet Austria [6]), credit institutions typically have up to 30 days to issue adverse action notices. In traditional underwriting, feature attributions are computed asynchronously in background batch jobs and archived in immutable compliance logs. For these workflows, latency is not a critical operational constraint, making exact Shapley values the unambiguous standard.
2) Synchronous Real-Time Inference: In high-frequency e-commerce checkout APIs, point-of-sale financing (Buy-Now-Pay-Later, BNPL), and interactive consumer counterfactual simulators (where users explore how adjusting credit utilization alters approval likelihood), explanations must be returned synchronously within microservice SLAs ($p95 < 50\,\text{ms}$). In this single-inference regime on standard low-dimensional credit datasets ($d = 10$), TreeSHAP executes in $\approx 1.5\,\text{ms}$ with a $> 30\times$ safety margin, demonstrating that operational trade-offs do not force compromises on faithfulness for single-query scoring.

B. Amortized Neural Approximation Risk and Legal Defensibility

While FastSHAP offers dramatic attribution acceleration ($27\,\mu\text{s}$), it replaces exact cooperative game calculations with a parameterized neural surrogate (ϕ_θ). Our empirical findings reveal two critical governance concerns:

1) Primary Adverse Action Misalignment: On XGBoost, while distilled FastSHAP achieves 83.3% Top-1 agreement on rejected applicants and 95.4% for decisive inputs, 16.7% of the Adverse Action Cohort receive a divergent primary reason code. While US ECOA Regulation B and CFPB Circular 2022-03 mitigate this by mandating disclosure of up to four principal reasons (where our Top-4 Jaccard reaches 77.7%), relying on single-factor Top-1 surrogate outputs introduces avoidable audit friction.
2) Off-Manifold Attribution Drift: Under $+3.0\,\sigma$ covariate shift, FastSHAP exhibits a severe collapse in rank correlation ($\rho_s = 0.116$). Furthermore, as demonstrated by Slack et al. [14], post-hoc perturbation explainers can be susceptible to adversarial shifts in off-manifold regions.

Because TreeSHAP executes in $\approx 1.5\,\text{ms}$ on GBDTs and Exact Shapley in 6.8 ms on MLPs, switching to FastSHAP saves merely 1.4–6.7 ms in single-query regimes while introducing surrogate drift risks.

C. Operational Decision Framework

We formulate a three-tier decision rubric for financial institutions:

1) Tier 1 (Tree Ensembles): Deploy TreeSHAP. It guarantees exact cooperative attributions relative to

the marginal expectation value function in ≈ 1.5 ms, exhibits perfect determinism across runs, and comfortably satisfies real-time SLAs.

2) Tier 2 (Deep Neural Architectures): In tabular neural models where TreeSHAP is inapplicable, deploy Exact Shapley for standard feature sets ($d \leq 12$, computed in < 28 ms via vectorized power-set masks on $B = 100$ background samples). For $d \leq 12$, Exact Shapley satisfies strict 50 ms SLAs while providing 100% mathematical faithfulness and zero surrogate drift. If FastSHAP is deployed in ultra-high-throughput streaming environments ($> 5,000$ QPS) or high dimensions ($d > 50$), train via teacher distillation and implement real-time covariate drift monitors to flag out-of-distribution inputs.
3) Tier 3 (Surrogate Governance): When deploying sampling explainers such as LIME, fix pseudo-random seeds and archive explanations synchronously at decision time to eliminate audit discrepancies arising from the 0.714 Top-3 Jaccard re-evaluation drift.

D. Scope of Mathematical Exactness

We emphasize that interventional TreeSHAP and our vectorized Exact Shapley solver are exact strictly with respect to the marginal expectation value function evaluated against empirical background samples ($v(S) = \mathbb{E}_{X_{\bar{S}}}[f(x_S, X_{\bar{S}})]$). This formulation evaluates feature subsets assuming marginal independence between S and $\bar{S}$, avoiding out-of-distribution conditioning artifacts while reflecting the empirical baseline distribution. It does not represent physical causal truth and should be interpreted as cooperative game-theoretic attribution.

VIII. Threats to Validity and Limitations

- Feature Dimensionality ($d = 10$ vs $d = 200$): Our primary empirical dataset reflects low-dimensional consumer credit scoring ($d = 10$). In enterprise banking environments with hundreds of transactional features ($d \geq 100$), TreeSHAP path complexity ($O(TLD^2)$) and KernelSHAP sampling scale significantly, creating true latency-faithfulness trade-offs as characterized in our scaling simulation.
- Greedy Deletion Proxy: Sequential feature deletion with median baselines evaluates sensitivity along a single elimination trajectory, which serves as an operational proxy with high variance rather than an axiomatic ground truth.
- Hardware Environment: Absolute latencies were measured on an Apple Silicon arm64 single-core environment; production cloud instances (e.g., x86_64 Kubernetes clusters) will exhibit proportional baseline shifts.

IX. Conclusion

This empirical benchmark systematically evaluated runtime latency, explanation faithfulness, stability, and amortization economics across 12 configurations in automated credit scoring. Our primary empirical finding is that on standard low-dimensional tabular features ($d = 10$), all four XAI paradigms satisfy 50 ms SLAs with 100% compliance, demonstrating that single-decision credit scoring does not face an operational latency-faithfulness bottleneck. Trade-offs emerge specifically in ultra-high-throughput streaming architectures ($> 5,000$ QPS) or high-dimensional feature spaces ($d > 50$). For tree ensembles, TreeSHAP provides exact cooperative attributions in ≈ 1.5 ms with zero surrogate drift. For neural architectures at $d \leq 12$, Exact Shapley runs in < 28 ms (6.78 ms at $d = 10$), providing zero-drift compliance within real-time SLAs. While amortized FastSHAP delivers a 22-fold end-to-end system speedup, its adverse action divergence on non-decisive inputs and susceptibility to OOD covariate shift require robust drift governance in regulated financial deployments.

Acknowledgment

In accordance with IEEE Authorship Policy, generative AI tools (Google DeepMind Gemini models) were utilized solely for code refactoring, language polishing, and LaTeX typesetting assistance. All conceptual frameworks, mathematical formulations, experimental pipelines, empirical data collection, statistical analyses, and scientific conclusions were conceived, executed, and validated independently by the author.

Reproducibility & Open Source Artifacts

All source code, configuration files (including random seeds), trained model weights, result tables, and experiment scripts are available in the accompanying repository: https://github.com/gadirlianar/credit-xai-benchmark. The raw dataset is not redistributed; it is obtained via the included download script under its original license.

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